

Compare the stemming and lemmatization results

Original	Stemming	Lemmatizing
businesses	busi	Businesses
strategies	strategi	strategy
regularly	regularli	regularly
this	thi	This
Online	onlin	online
compare	compar	compare
purchase	purchas	purchase
Provide	provid	provide
Increase	increas	increase
has	ha	ha
customers	custom	Customers

The above table is an AI generated representation including all the words that changed after stemming and lemmatizing
(The original form of the data was fed into gemini along with the results from stemming and lemmatizing. Then it was prompted to create a table consisting of all the provided data for a clear comparison)

	Represents all occasions where the original meaning of the word is retained
	Represents all occasions where the original meaning of the word is not at all retained
	Represents all occasions where the words were very minimally altered

It can be clearly seen that lemmatizing was able to retain the meanings of almost all the original words in comparison to stemming where there are a lot of mistakes.

Both lemmatizing and stemming have struggled with the words 'has' and 'strategies' but overall lemmatizing clearly has a better result.

Briefly explain the difference between stemming and lemmatization

Stemming works with a set of rules and removes parts of words using those rules.

Lemmatization however works with a different approach. It uses a dictionary and strict grammar rules to obtain a clear base word.

Stemming is comparatively much faster than lemmatization but that speed comes at the cost of accuracy.